CHAYA छाया · system architecture

Two data streams in, one calibrated virtual patient; 2-hour spike and low-glucose warnings, what-if simulation and a patient app out. Patient logs flow back in.

1 · DATA STREAMS

Static / historical: EHR

FHIR R4 Bundle per patient

- Demographics, city, diet
- Diagnoses (SNOMED): T2D, HTN, NAFLD
- Labs (LOINC): HbA1c, FPG, lipids, eGFR
- Medications: metformin, glimepiride, DPP-4i, SGLT2i, insulin
- Genetics: TCF7L2 rs7903146

Dynamic / real-time: wearables

- CGM glucose, every 5 min
- Steps, heart rate
- Nightly HRV (RMSSD), sleep stages
- Meal and dose logs, now from the patient app

Synthetic cohort

150 Indian patients × 14 days · 604,800 sensor rows · physiology-driven generator. No real patient data.

2 · INGEST & HARMONISE

FHIR parser

LOINC / SNOMED codes → patient profile vector

Stream alignment

5-min grid · sensor gaps kept as gaps · meal and dose events placed on the timeline

Model inputs

Minute-level carbs, GI-dependent absorption, steps, sleep-debt and missed-dose modifiers

3 · DIGITAL TWIN CORE

Per-patient physiological model

`gut → plasma glucose ⇄ insulin action`
`extended Bergman minimal model`

1. **EHR prior.** BMI, waist, HbA1c, duration, TCF7L2, drugs → starting physiology
2. **Calibrate.** MAP fit to 7 days of CGM: insulin sensitivity, β-cell response, basal glucose, meal bioavailability, dawn effect
3. **Observe.** Runs beside live CGM; corrects its state and detects unlogged meals
4. **Habit model.** Learns when and how much this patient usually eats
5. **Forecast & what-if.** Simulates the next 2–4 h under routine or a doctor's scenario

↻ re-calibrates as new data arrives

4 · FUSION & PREDICTION

Fused feature vector

every 15 min

`18 EHR` `25 wearable` `11 twin`

Spike classifier

LightGBM · P(glucose > 180 mg/dL within 2 h)

Glucose regressor

LightGBM · glucose value at +2 h

Low-glucose risk NEW

Twin forecast < 80 mg/dL in 2 h → *Low likely*; CGM < 70 → *Low now*. 7 of 10 lows warned, median 80 min ahead

Explanations

Tree SHAP → clinical drivers: meal, routine, sleep, activity, medication, physiology

5 · DELIVERY

Simulated clinic

- Pinned register, case sheet, stamps
- Reasons, twin physiology
- Escalation: owner, WhatsApp, call log
- Rx pad dock: what-if → save & send
- Alert threshold · Outcomes page

Patient app NEW

- English · हिंदी
- Glucose + 2 h shadow, plain warnings
- Dose reminders, meal logging
- Doctor's advice with read receipt

↺ meals and doses logged in the app feed the twin's inputs

Evaluation protocol

Patient-level split 105 train / 45 held-out. Temporal split: days 1–7 calibrate, days 8–14 predict. Threshold chosen on training folds only.

Privacy by design

Synthetic data only (DPDP Act 2023 / HIPAA sandbox rules). FHIR-native so it can plug into ABDM-consented records later.

Stack

Python · NumPy/Numba · SciPy · LightGBM · FastAPI (local pipeline) · Vercel functions (Mumbai) · Supabase Postgres · vanilla JS + D3

From a warning to an action the patient sees

What happens after the twin raises a warning, where each step is stored, and how the live prototype is deployed

THE CLOSED CARE LOOP

1 · warn
Twin + model
Spike likely or Low likely, 2 h ahead, with three reasons. Threshold set by the clinic (2.6–4.6 warnings per patient a day).

2 · review
Clinician
Reads the stamp and reasons; marks the warning reviewed.

`alert_reviews`

3 · escalate
Owner + due time
Care coordinator, doctor on call or educator. Prefilled WhatsApp or a call log; overdue moves to the doctor on call.

`escalations`

4 · advise
Rx pad
Test a meal, walk, sleep or missed dose on the twin, then save and send.

`case_notes`

↓ then on the patient's phone, and back to the clinic

5 · reach
Patient app
Advice arrives in the doctor's hand with the peak it should prevent. A toast in the clinic links straight to it.

6 · read
Read receipt
Patient taps Got it; the clinic sees when.

`case_notes.ack_at`

7 · log
Meals and doses
Logged in two taps; land on the clinic's day trace and the twin's inputs.

`patient_logs`

8 · measure
Clinic outcomes
Time in range, low safety, alert load and the funnel warnings → reviewed → escalated → contacted → advice → seen.

WHAT IS NEW IN THIS VERSION

Low-glucose warnings
Same twin, read downwards; shown first on the patient's phone with the 15 g sugar rule.

Escalation and outcomes
Every warning has an owner; the buyer sees whether warnings became action.

Bilingual patient app
English and Hindi, dose reminders, logging, advice read receipts, its own 11-step tour.

Clinic workspace
Panes pinned to the screen edges, Rx pad as a compose-style dock, clear home / clinic / patient navigation, 25 data sources documented.

LIVE DEPLOYMENT

Browser
Static app · D3 · twin simulator in JS (matches Python to 0.1 mg/dL)

→

Vercel API
Node function, Mumbai (bom1)

→

Supabase
Postgres, row-level security, column-level write grants

Tables

`patients`	register: 45 synthetic patients, first name + surname
`cases`	replay week per patient as JSONB: CGM, meals, predictions, twin state
`case_notes`	Rx advice and its projection; ack_at when read
`alert_reviews`	warnings marked reviewed
`escalations`	owner, due time, channel, outcome, status
`patient_logs`	meals and doses from the patient app
`meta`	threshold, metrics, meal catalogue, population

Offline fallback
If the API is unreachable after a cold-start wait and one retry, the app says *Offline demo* and runs from bundled data; changes stay in the browser.

Offline compute
`run_all.sh`: data → twins → LightGBM → metrics → web data. `build_outcomes.js`: low flags, threshold curve, daily time in range, using the browser's own simulator.